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Discussion Agenda 08/17/2013

Below is a discussion agenda in an approximate order of importance for the 2nd AGORA Workshop @ University of California Santa Cruz (Aug. 16–18; see the [program](#))

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[1. The Flagship Paper](#)

– The AGORA Flagship paper has been written thanks in great part to the contributions by many of you, and is now submitted to the ApJS. Please see [this page](#) for a local draft in high-resolution and other quick links.

– We thank all the people who have contributed to the writing of this article. As for the authorship of the paper, please see the guideline discussed [before](#). The author list currently includes **only** the individuals who have explicitly contributed to the text of this article, individuals who have made contributions in WGs I–IV (in consultation with WG leaders), individuals who have run the proof-of-concept test, and members of the steering committee and the NHiRGs team. The list will eventually encompass "all the people who have made meaningful contributions to the project, not just to this paper, up to the point of the submission of the article." The current authorship and its order have been reviewed and decided by the steering committee as of Aug. 6, 2013.

[2. Discussions Report](#)

The following table summarizes the result of discussions and/or progress made so far in each Working Group (updated by Working Group leaders based on the plenary and breakout discussion sessions on 08/17–18/2013).

	WORKING GROUP	SUMMARY OF DISCUSSION / PROGRESS	
I	Common Physics and Introduction to Project (discussion led by Madau)	– Who will be in charge of implementing and testing the common physics package in each code? Then each code group internally decides who implements which.	
		Code	Contacts
		ART-I (NMSU)	Daniel Ceverino
		ART-II (Chicago)	Sam Leitner
		ENZO	Ji-hoon Kim
		GADGET	Alex Hobbs, Ken Nagamine
		GASOLINE	Sijing Shen
		RAMSES	Oscar Agertz
		– Recommended resolution: 100 pc or better (~Eris; at least for 1e11 Ms halos), to make common physics and SF/Fbck prescriptions reasonable and reliable; but may vary depending on problem types one investigates. See here for reference.	
		– Future milestone: We aim to have a working version of each code with the common physics package (not SF/Fbck) by the next web conference. We will then consider testing out the codes by borrowing a setup of the Proof-of-concept-NOSUBGRID run (see WGs III+VI below).	

II+V	Common ICs: Isolated + Isolated Galaxies and Subgrid Physics (talk by Agertz; discussion led by Wadsley)	Code ART-I (NMSU) ART-II (Chicago) ENZO GADGET GASOLINE RAMSES Contacts Anatoly Klypin Sam Leitner Nathan Goldbaum Alex Hobbs, Ken Nagamine Sijing Shen, James Wadsley Oscar Agertz, Romain Teyssier – Please see the detailed goals of these Working Groups, and how to calibrate SF/Fbck prescriptions across codes at the WG V page. – Future milestone: (to be further discussed)
III+VI	Common ICs: Cosmological (talk by Hahn) + Dwarf Galaxies in Cosmological Simulations (talk by Onorbe)	[Effort to push out PAPER "2" using Proof-of-concept-DM run (~300 / ~100 pc resolution), led by Oliver Hahn] – Comparison to be done: mass accretion, profiles at various redshifts, early formation history properties, etc. – Include multi-mass Rockstar in yt, switch from subhalo mass to V_max – Future milestone: Definitions of highest-resolution regions and masks will be provided by Sep. 13 with initial testing for 2+ codes, and with a point-release of MUSIC everyone should use. Supports for refinement masks in all codes need to be coded up by then. After that low-res and hi-res Proof-of-concept-DM need to be rerun by all codes using the same highest-resolution regions. Support for Rockstar in yt for all codes should be available by the same time. First analysis will follow after that as simulation snapshots come in. First results will be discussed as available in the next web conference. [Effort to push out PAPER "3" using Proof-of-concept-NOSUBGRID run (~500 pc resolution), led by Nick Gnedin] – Step-by-step approach to start hydrodynamics production runs {Step 1} Dark matter-only run, if not done yet {Step 2} Hydrodynamic run with ~500 pc to ~1 kpc resolution + Vanilla physics (including photo-ionizing UV background, Grackle radiative cooling, adiabatic EoS above n_{crit}, but no subgrid physics, i.e. no star formation or feedback) {Step 3} The actual production run with ~100 pc resolution + common physics package + common parameter choices – Who is going to run Proof-of-concept-NOSUBGRID (~1e11 Ms quiescent halo) for each code? Code ART-I ART-II CHANGA ENZO GADGET GASOLINE PKDGRAV-2 RAMSES Proof-of-concept-DM (Klypin) Leitner Quinn Kim Hahn/Todoroki/Hobbs Shen Guedes/Kuhlen Teyssier Proof-of-concept-NOSUBGRID Ceverino/Klypin Leitner (Quinn) Hummels/Kim/Wise Thompson Shen N/A Agertz/Teyssier
		– We will make agora-analysis-script repository to be the place where "direct technology transfer" of common analysis occurs within and across code platforms

IV	Common Analysis (talk by Turk)	for AGORA. All participants of the Proof-of-concept-DM/NOSUBGRID runs now have read/write accesses to contribute new technology. – Milestones: The status of yt-3.0 is that currently the Octree and Patch codes used in the Flagship paper are all supported. For the particle/N-body codes, access of particles and indexing of those particles is supported, as well as first stage visualizations. See the release note of yt-3.0a3. SPH smoothing is the next stage, as well as improving reliability and error resiliency of the particle frontends. – See also the other yt-related talks: (1) yt for AGORA (I) – Live demos of practical yt usages by Nathan Goldbaum (2) yt for AGORA (II) – How to get help by Nathan Goldbaum (3) yt+Sunrise for AGORA by Chris Moody
VII+VIII	Dark Matter (talk by Rocha) + Satellite Galaxies (talk by Zolotov)	– Please see Working Group VII: Dark Matter Workspace
IX+XII	Characteristics of Cosmological Galaxies (talk by Guedes & Hummels) + Interstellar Medium (talk by Leitner)	– For now, members are encouraged to participate in Paper "3", implement the common physics package, and run the isolated disk simulation. A web conference consisting of simulators will be organized in several weeks to gather progress updates on these tasks from each group, and make concrete plans for the cooling+SF/Fbck runs.
X+XI	Outflows (talk by Shen) + High-redshift Galaxies (talk by Ceverino)	– Two issues were raised during the discussion: – First, to study CGM, we agreed that it is desirable to have the high resolution region to be 3 R_{vir}, but if that is difficult, it should be *at least* 300 kpc from the center of the main galaxy. The zoom-in regions in the current ICs are 2 R_{vir}. For halos larger than $1e11$ solar mass, 2 R_{vir} is larger than 300 kpc. For dwarf galaxies, it will be great to increase to 3–4 R_{vir}. But tests need to be done to find the balance between resolution requirement and the zoom-in size. It remains to be discussed and decided before the runs (especially hydro runs) started. – Second, Michele Fumagalli and Sebastiano Cantalupo raise the question of whether we want to include a certain self-shielding approximation (e.g. turn off the UV when density $> 0.01 - 0.1 \text{ cm}^{-3}$) in the runs. This is not hard to implement, and the cooling with no UVB is handled in Grackle. But it may require some test (maybe in isolated galaxy runs). We'd like to know what everyone's opinion on this.
XIII	Black Hole Accretion and Feedback (talk by Hobbs)	– The main action item is to decide upon a BH accretion model. The different models are listed in my talk (along with references) and I also encourage everyone to read the following papers in order to make an informed choice: [1], [2], [3] – The model does not have to be the same for both types of simulation (isolated discs and cosmo runs; indeed, in my talk I argue for different models due to the differences in resolution. We will take a vote on this and discuss the other open questions + action items in a SeeVogh meeting in September – email with Doodle poll to follow.

[3. Live AGORA Simulations Tracker](#)

- Let us keep track of the available simulation outputs in the Collaboration so that science-oriented WGs can better coordinate. Please check out [THIS PAGE](#) where we intend to collect such information. (Thanks to Annalisa Pillepich and Dylan Tweed; currently under construction)
- Who are going to run which of the 8 cosmological halos for each code group, starting with DM-only runs? Can each group afford to do what in practical terms?

Code	1e10q	1e10v	1e11q	1e11v	1e12q	1e12v	1e13q	1e13v
ART-I			DC					
ART-II			SL		SL			
ENZO	CS	CS	JK					
GADGET-2/3 (various versions)	JO/MR	JO/MR	MR/AH/KT	MR	MR	MR		
GASOLINE			SS				SS	SS
PKDGRAV-2			JG/MK					
RAMSES			RT					

[4. How to Acknowledge AGORA in Private Research](#)

- Can we use aspects of AGORA for a private research project (i.e. the common ICs, common physics, etc.) and subsequently publish it separately?
- [\[Proposed Policy\]](#)
As soon as the AGORA Flagship paper is submitted and published, the common ICs and the common physics package belong to the public domain. Therefore, everyone – regardless of his/her affiliation with the AGORA Collaboration – should be able to use the AGORA ICs and physics packages in his/her private research projects. The AGORA flagship paper should be cited, however, and an acknowledgment would be nice.
- General authorship policy in AGORA: please check the previous [discussion](#)

[5. How Often Do We Store the Data?](#)

- Minimum requirement for data storage for each code's production runs: 400 timesteps (increased from 200) equally spaced in expansion parameter in addition to redshift snapshots at $z = 6, 3, 2, 1, 0.5, 0.2$, and 0.0 at the very least. This would mean $400 * \sim 1 \text{ GB} = \sim 400 \text{ GB}$ per simulation. --> **Outdated.** See the new strategy [here](#).

[6. Future Project Timelines, etc.](#)

- Next Web Conference (tentative): mid-November, 2013
- Computing time proposal by the AGORA Collaboration as a whole: to be discussed by the Steering Committee, this planned proposal does not prevent individual members from submitting individual computing proposals related to the AGORA research.
- Plan for the central data repository using the Huawei storage system@UCSC: system configuration going on as of Jul. 2013, more logistics to be decided before we can give Hyades accounts to AGORA participants, won't be available at least in the next few months

